```
2091  0  INSERT FILE='E:/Documents/GitHub/biostatistics-portfolio/_spss_tmp/hwg-ttest-cor
relation.sps' ERROR=CONTINUE.
2091  0  * Set once: the folder that contains this repository, then run.
2092  0  CD 'E:/Documents/GitHub/biostatistics-portfolio'.
2093  0  GET FILE='data/raw/hwg.sav'.
2094  0  DATASET NAME DataSet0 WINDOW=FRONT.
2095  0  * Encoding: UTF-8.
2096  0
2097  0  DATASET ACTIVATE DataSet0.
2098  0  FREQUENCIES VARIABLES=Gender Height Weight
2099  0    /ORDER=ANALYSIS.
```

Frequencies

Notes

Output Created		03-NOV-2026 15:38:14
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Gender Height Weight /ORDER=ANALYSIS.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.03

[DataSet0] E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav

Statistics

		Gender	Height in inches	Weight in pounds
N	Valid	12	10	10
	Missing	0	2	2

Frequency Table

Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	5	41.7	41.7	41.7
	Female	7	58.3	58.3	100.0
	Total	12	100.0	100.0	

Height in inches

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	62.00	1	8.3	10.0	10.0
	64.00	3	25.0	30.0	40.0
	66.00	3	25.0	30.0	70.0
	68.00	2	16.7	20.0	90.0
	70.00	1	8.3	10.0	100.0
	Total	10	83.3	100.0	
Missing	System	2	16.7		
Total		12	100.0		

Weight in pounds

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	116.00	1	8.3	10.0	10.0
	118.00	1	8.3	10.0	20.0
	122.00	1	8.3	10.0	30.0
	124.00	1	8.3	10.0	40.0
	128.00	1	8.3	10.0	50.0
	136.00	1	8.3	10.0	60.0
	138.00	1	8.3	10.0	70.0
	144.00	1	8.3	10.0	80.0
	151.00	1	8.3	10.0	90.0
	153.00	1	8.3	10.0	100.0
	Total	10	83.3	100.0	
Missing	System	2	16.7		
Total		12	100.0		

```
2100  0
2101  0  FREQUENCIES VARIABLES=Gender Height Weight
2102  0    /FORMAT=NOTABLE
2103  0    /BARCHART PERCENT
2104  0    /ORDER=ANALYSIS.
```

Frequencies

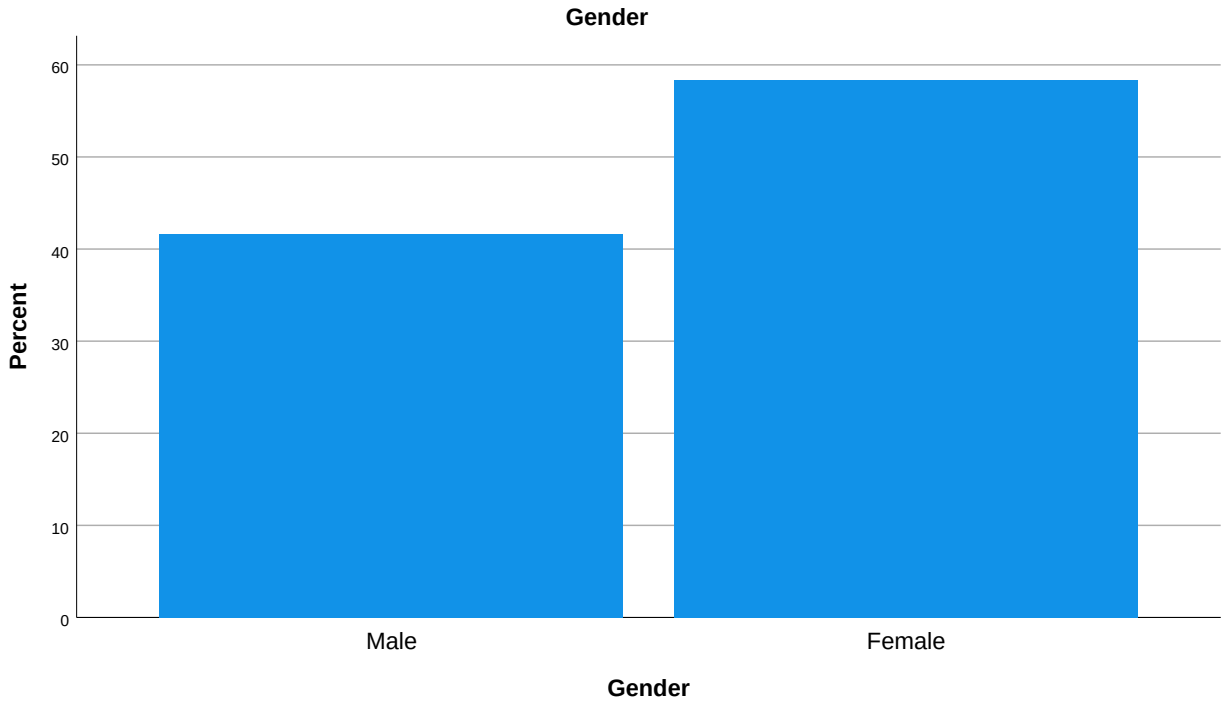
Notes

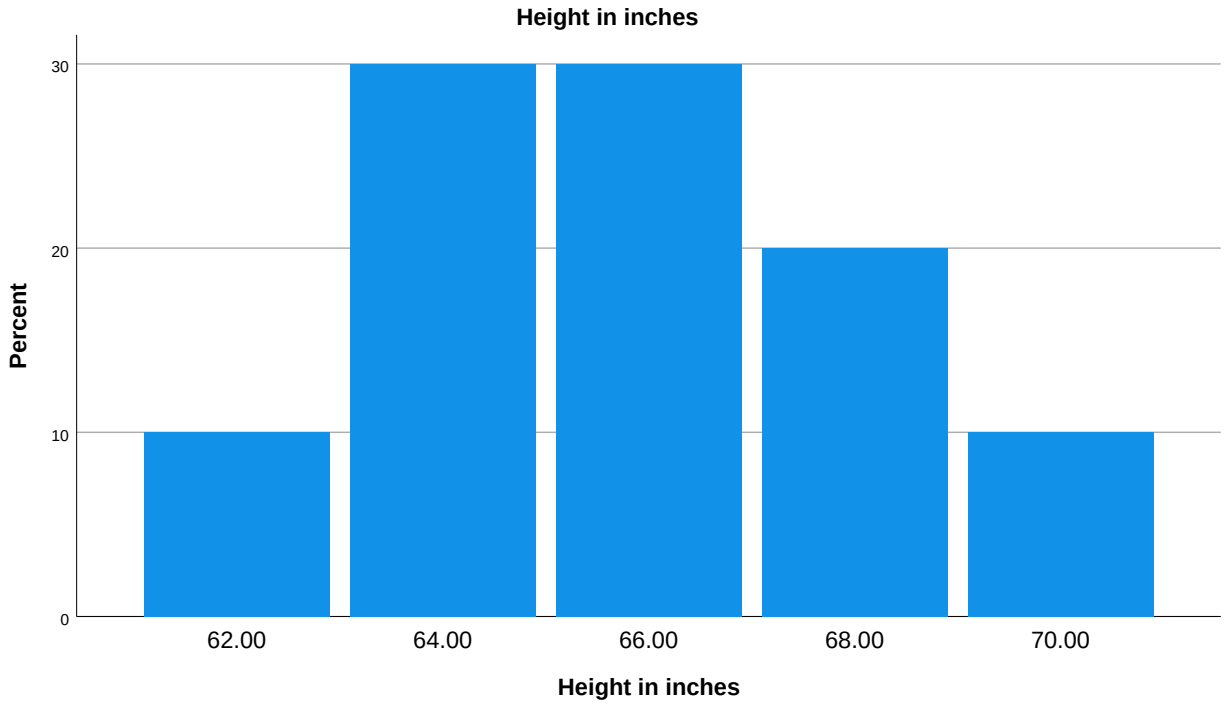
Output Created		03-NOV-2026 15:38:14
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Gender Height Weight /FORMAT=NOTABLE /BARCHART PERCENT /ORDER=ANALYSIS.
Resources	Processor Time	00:00:01.28
	Elapsed Time	00:00:01.91

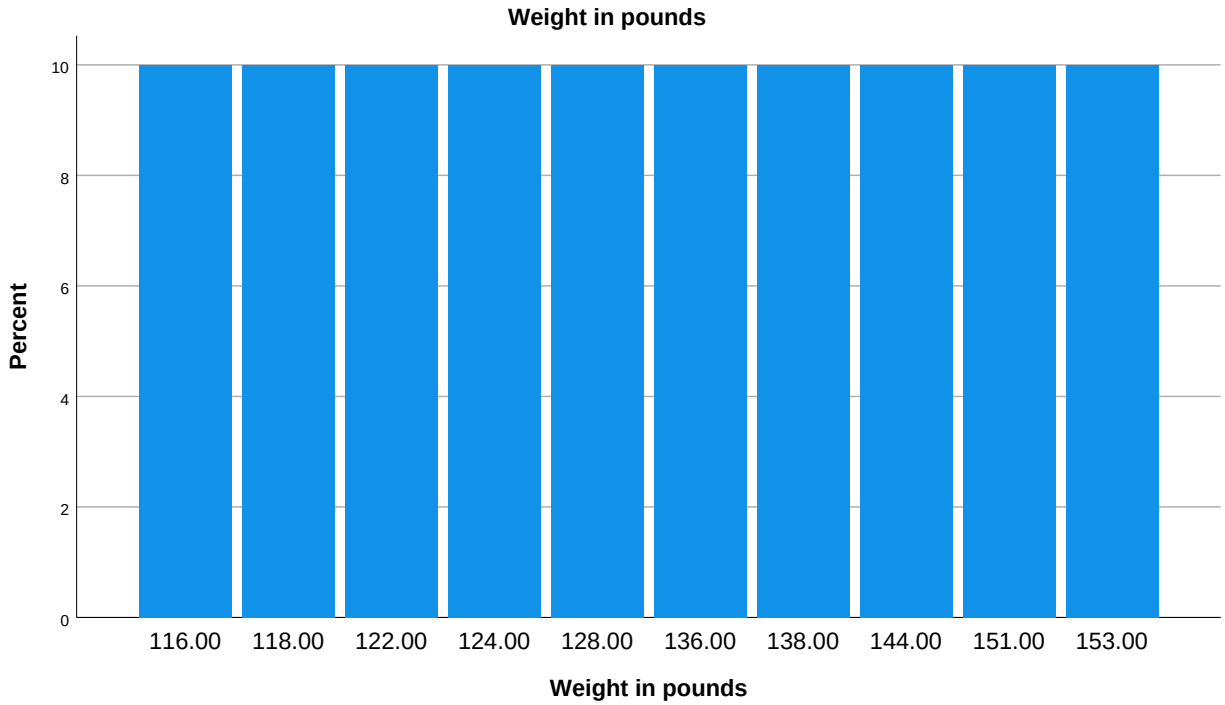
Statistics

		Gender	Height in inches	Weight in pounds
N	Valid	12	10	10
	Missing	0	2	2

Bar Chart


Gender
Percent
60
50
40
30
20
10
0
Male
Female
Gender


Height in inches
Percent
30
20
10
0
62.00
64.00
66.00
68.00
70.00
Height in inches




```
2105  0
2106  0  FREQUENCIES VARIABLES=Gender Height Weight
2107  0    /FORMAT=NOTABLE
2108  0    /STATISTICS=STDDEV VARIANCE MINIMUM MAXIMUM SEMEAN MEAN
2109  0    /HISTOGRAM NORMAL
2110  0    /ORDER=ANALYSIS.
```

Frequencies

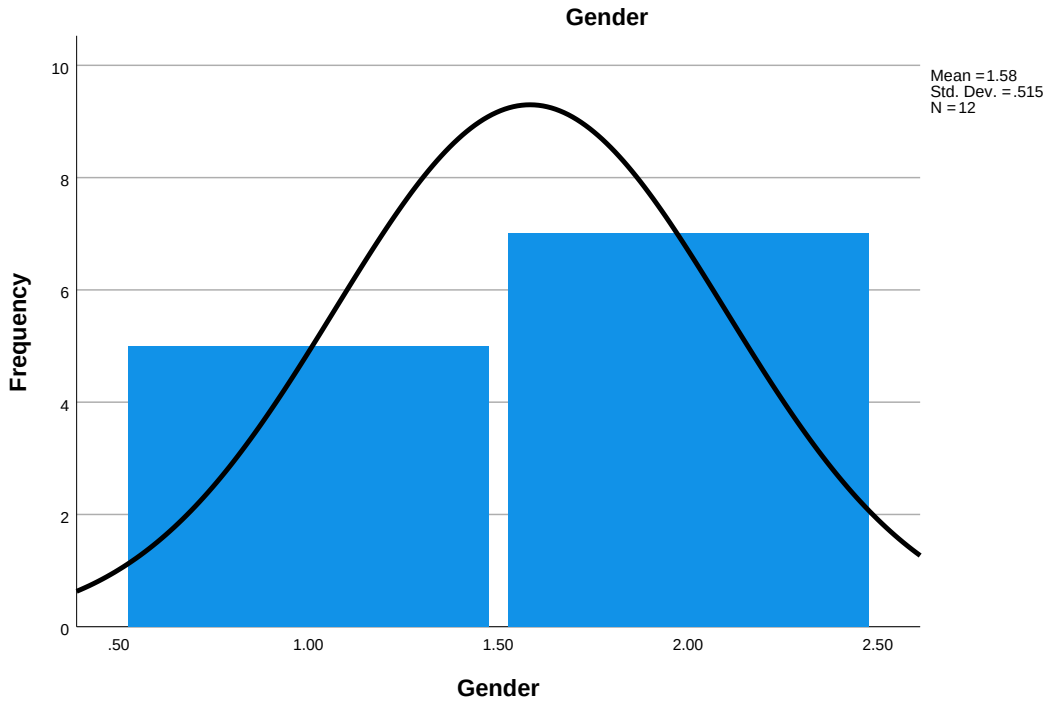
Notes

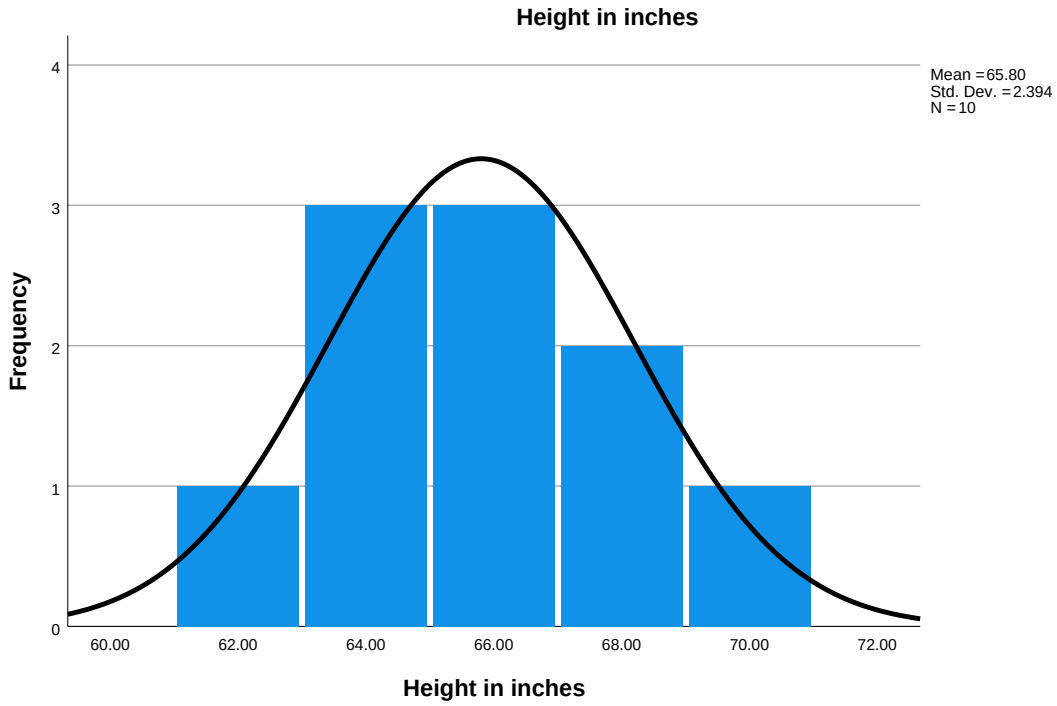
Output Created		03-NOV-2026 15:38:16
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Gender Height Weight /FORMAT=NOTABLE /STATISTICS=STDDEV VARIANCE MINIMUM MAXIMUM SEMEAN MEAN /HISTOGRAM NORMAL...
Resources	Processor Time	00:00:01.16
	Elapsed Time	00:00:01.40

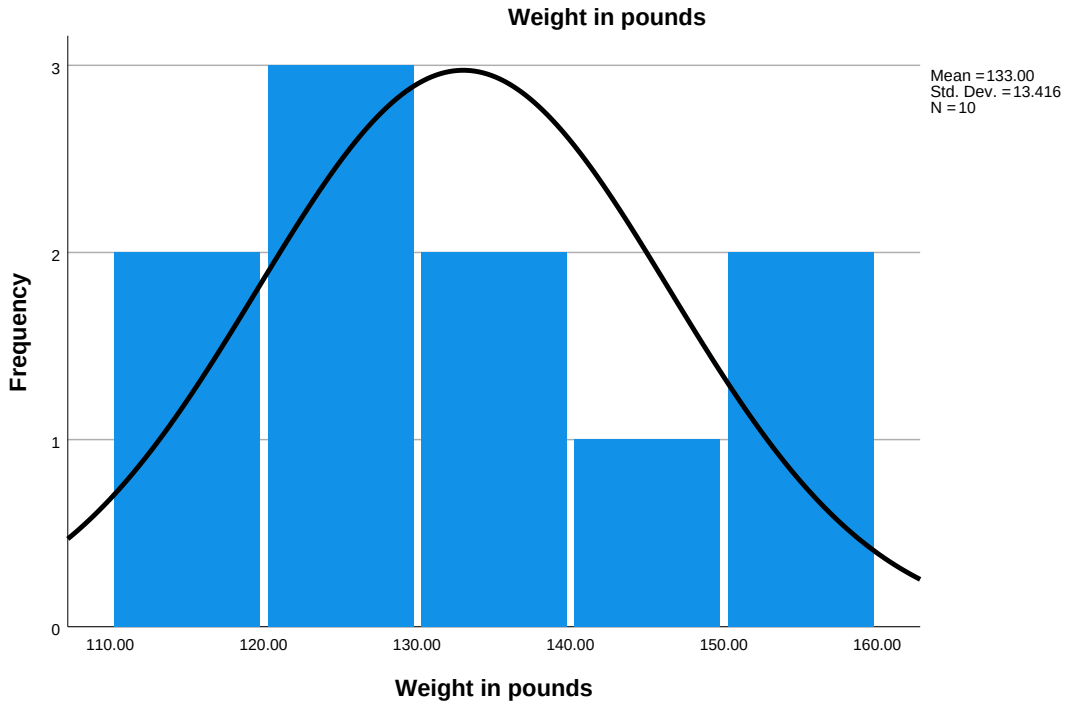
Statistics

		Gender	Height in inches	Weight in pounds
N	Valid	12	10	10
	Missing	0	2	2
Mean		1.5833	65.8000	133.0000
Std. Error of Mean		.14865	.75719	4.24264
Std. Deviation		.51493	2.39444	13.41641
Variance		.265	5.733	180.000
Minimum		1.00	62.00	116.00
Maximum		2.00	70.00	153.00

Histogram


Gender
Mean = 1.58
Std. Dev. = .515
N = 12
Frequency
10
8
6
4
2
0
.50
1.00
1.50
2.00
2.50
Gender


Height in inches
Mean = 65.80
Std. Dev. = 2.394
N = 10
Frequency
4
3
2
1
0
60.00
62.00
64.00
66.00
68.00
70.00
72.00
Height in inches




```
2111  0
2112  0  MEANS TABLES=Height BY Gender
2113  0    /CELLS=MEAN COUNT STDDEV.
```

Means

Notes

Output Created		03-NOV-2026 15:38:17
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	For each dependent variable in a table, user-defined missing values for the dependent and all grouping variables are treated as missing.
	Cases Used	Cases used for each table have no missing values in any independent variable, and not all dependent variables have missing values.
Syntax		MEANS TABLES=Height BY Gender /CELLS=MEAN COUNT STDDEV.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

Case Processing Summary

	Cases					
	Included		Excluded		Total	
	N	Percent	N	Percent	N	Percent
Height in inches * Gender	10	83.3%	2	16.7%	12	100.0%

Report

Height in inches

Gender	Mean	N	Std. Deviation
Male	67.2000	5	2.28035
Female	64.4000	5	1.67332
Total	65.8000	10	2.39444

```
2114  0
2115  0  DESCRIPTIVES VARIABLES=Gender Height Weight
2116  0    /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Notes

Output Created		03-NOV-2026 15:38:17
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Gender Height Weight /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Gender	12	1.00	2.00	1.5833	.51493
Height in inches	10	62.00	70.00	65.8000	2.39444
Weight in pounds	10	116.00	153.00	133.0000	13.41641
Valid N (listwise)	10				

```
2117  0
2118  0  EXAMINE VARIABLES=Height Weight
2119  0    /PLOT BOXPLOT STEMLEAF HISTOGRAM
2120  0    /COMPARE GROUPS
2121  0    /PERCENTILES(5,10,25,50,75,90,95) HAVERAGE
2122  0    /STATISTICS DESCRIPTIVES EXTREME
2123  0    /CINTERVAL 95
2124  0    /MISSING LISTWISE
2125  0    /NOTOTAL.
```

Explore

Notes

Output Created		03-NOV-2026 15:38:17
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values for dependent variables are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.

Notes

Syntax		EXAMINE VARIABLES=Height Weight /PLOT BOXPLOT STEMLEAF HISTOGRAM /COMPARE GROUPS /PERCENTILES (5,10,25,50,75,90,95) HAVERAGE /STATISTICS DESCRIPTIVES EXTREME /CINTERVAL 95 /MISSING LISTWISE /NOTOTAL.
Resources	Processor Time	00:00:01.13
	Elapsed Time	00:00:01.36

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Height in inches	10	83.3%	2	16.7%	12	100.0%
Weight in pounds	10	83.3%	2	16.7%	12	100.0%

Descriptives

			Statistic	Std. Error
Height in inches	Mean		65.8000	.75719
	95% Confidence Interval for Mean	Lower Bound	64.0871	
		Upper Bound	67.5129	
	5% Trimmed Mean		65.7778	
	Median		66.0000	
	Variance		5.733	
	Std. Deviation		2.39444	
	Minimum		62.00	
	Maximum		70.00	
	Range		8.00	
	Interquartile Range		4.00	
	Skewness		.233	.687
	Kurtosis		-.369	1.334
Weight in pounds	Mean		133.0000	4.24264
	95% Confidence Interval for Mean	Lower Bound	123.4025	
		Upper Bound	142.5975	
	5% Trimmed Mean		132.8333	
	Median		132.0000	
	Variance		180.000	
	Std. Deviation		13.41641	
	Minimum		116.00	
	Maximum		153.00	
	Range		37.00	
	Interquartile Range		24.75	
	Skewness		.278	.687
	Kurtosis		-1.400	1.334

Percentiles

		Percentiles			
		5	10	25	50
Weighted Average (Definition 1)	Height in inches	62.0000	62.2000	64.0000	66.0000
	Weight in pounds	116.0000	116.2000	121.0000	132.0000
Tukey's Hinges	Height in inches			64.0000	66.0000
	Weight in pounds			122.0000	132.0000

Percentiles

		Percentiles		
		75	90	95
Weighted Average (Definition 1)	Height in inches	68.0000	69.8000	.
	Weight in pounds	145.7500	152.8000	.
Tukey's Hinges	Height in inches	68.0000		
	Weight in pounds	144.0000		

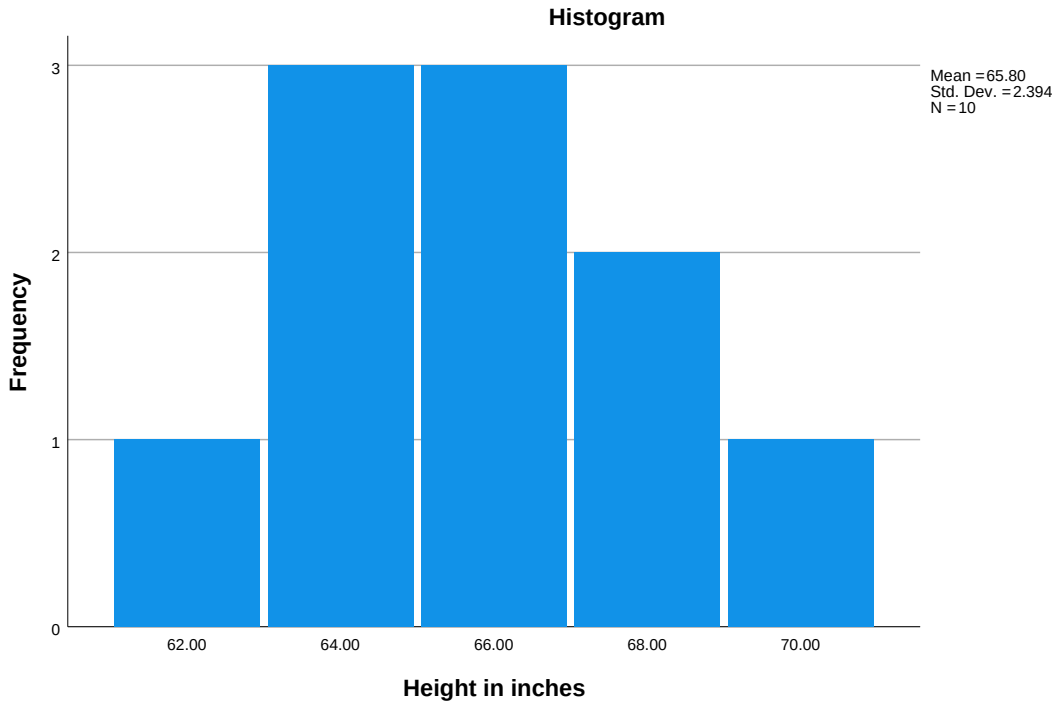
Extreme Values

			Case Number	Value
Height in inches	Highest	1	4	70.00
		2	5	68.00
		3	8	68.00
		4	2	66.00
		5	9	66.00[a]
	Lowest	1	3	62.00
		2	7	64.00
		3	6	64.00
		4	1	64.00
		5	10	66.00[b]
Weight in pounds	Highest	1	4	153.00
		2	9	151.00
		3	5	144.00
		4	8	138.00
		5	3	136.00
	Lowest	1	6	116.00
		2	10	118.00
		3	7	122.00
		4	2	124.00
		5	1	128.00

a. Only a partial list of cases with the value 66.00 are shown in the table of upper extremes.

b. Only a partial list of cases with the value 66.00 are shown in the table of lower extremes.

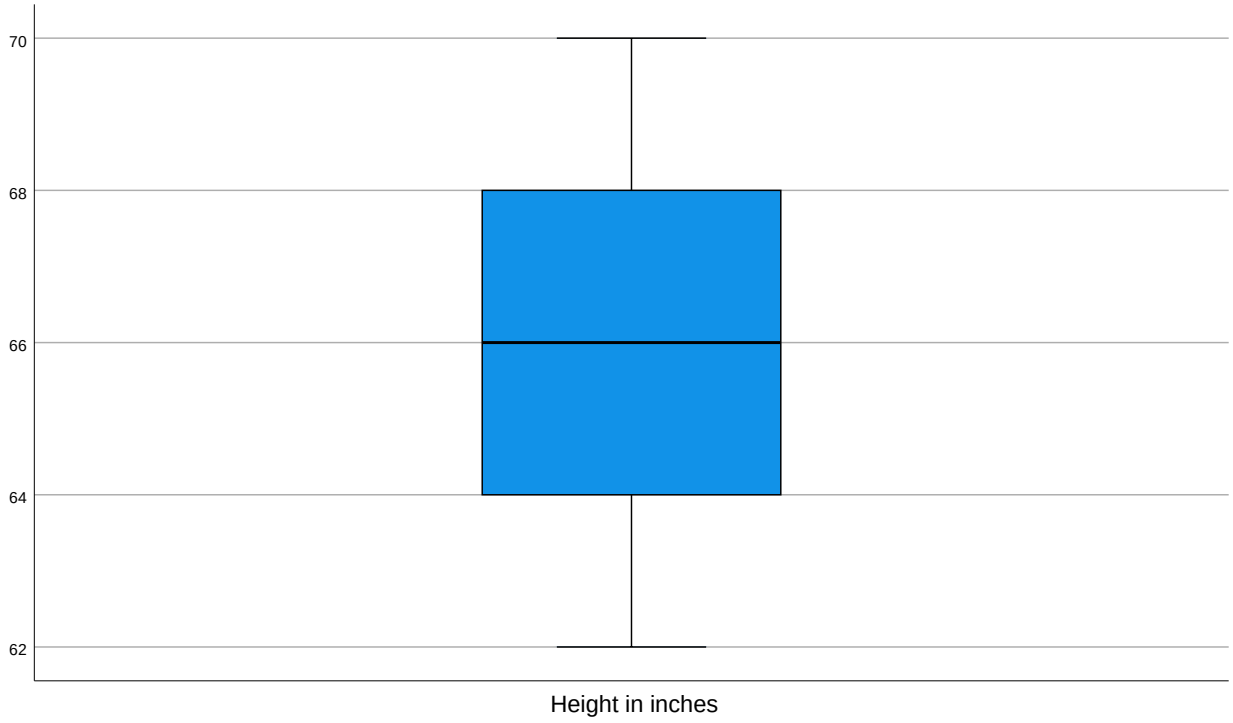
Height in inches



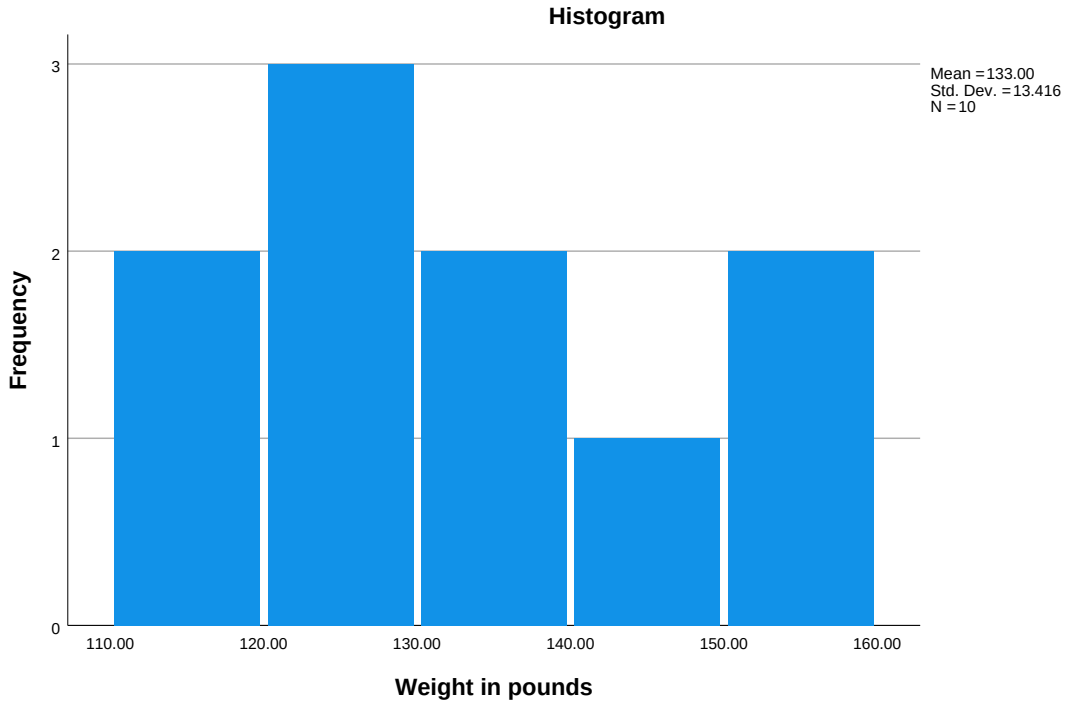
Height in inches Stem-and-Leaf Plot

Frequency	Stem &	Leaf
4.00	6 .	2444
5.00	6 .	66688
1.00	7 .	0

Stem width: 10.00
Each leaf: 1 case(s)



Weight in pounds

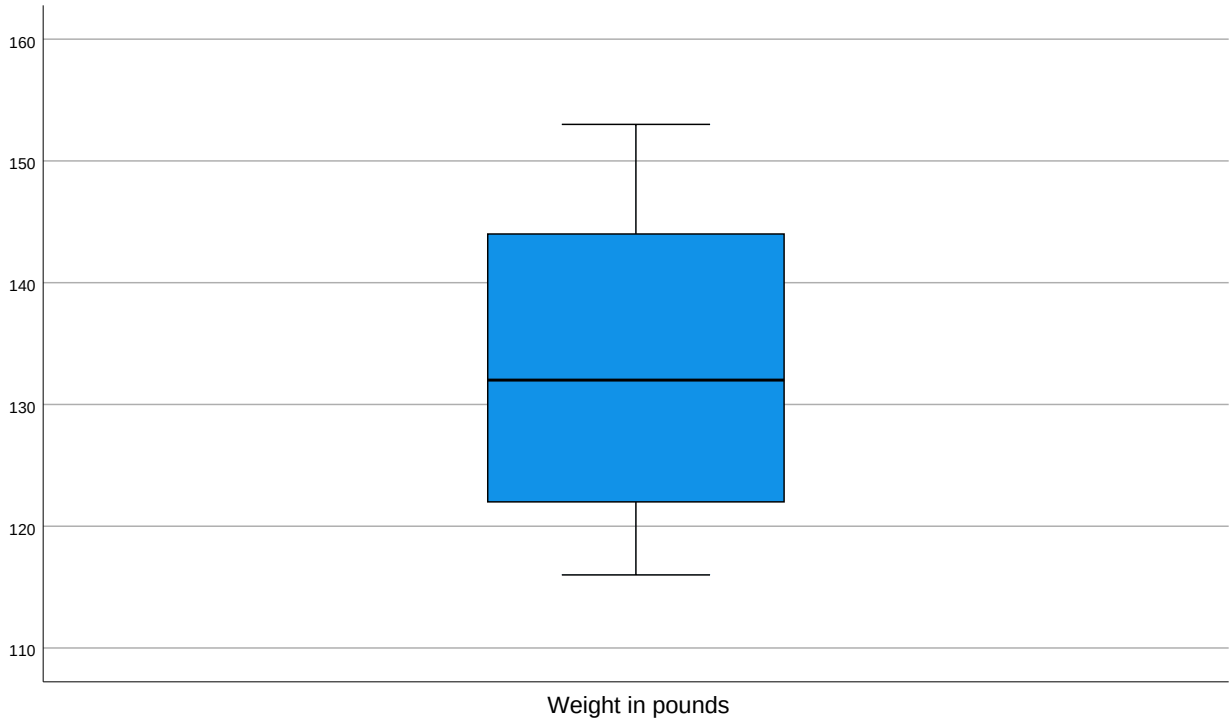



```
Weight in pounds Stem-and-Leaf Plot

 Frequency    Stem &  Leaf
```

```
    2.00       11 .  68
    3.00       12 .  248
    2.00       13 .  68
    1.00       14 .  4
    2.00       15 .  13

Stem width:     10.00
Each leaf:        1 case(s)
```




```
2126  0
2127  0  EXAMINE VARIABLES=Height Weight BY Gender
2128  0    /PLOT BOXPLOT STEMLEAF HISTOGRAM
2129  0    /COMPARE GROUPS
2130  0    /PERCENTILES(5,10,25,50,75,90,95) HAVERAGE
2131  0    /STATISTICS DESCRIPTIVES EXTREME
2132  0    /CINTERVAL 95
2133  0    /MISSING LISTWISE
2134  0    /NOTOTAL.
```

Explore

Notes

Output Created		03-NOV-2026 15:38:19
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values for dependent variables are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any dependent variable or factor used.
Syntax		EXAMINE VARIABLES=Height Weight BY Gender /PLOT BOXPLOT STEMLEAF HISTOGRAM /COMPARE GROUPS /PERCENTILES (5,10,25,50,75,90,95) HAVERAGE /STATISTICS DESCRIPTIVES EXTREME /CINTERVAL 95 /MISSING LISTWISE /NOTOTAL.
Resources	Processor Time	00:00:01.72
	Elapsed Time	00:00:01.85

Gender

Case Processing Summary

		Cases					
		Valid		Missing		Total	
	Gender	N	Percent	N	Percent	N	Percent
Height in inches	Male	5	100.0%	0	0.0%	5	100.0%
	Female	5	71.4%	2	28.6%	7	100.0%
Weight in pounds	Male	5	100.0%	0	0.0%	5	100.0%
	Female	5	71.4%	2	28.6%	7	100.0%

Descriptives

	Gender			Statistic	Std. Error
Height in inches	Male	Mean		67.2000	1.01980
		95% Confidence Interval for Mean	Lower Bound	64.3686	
			Upper Bound	70.0314	
		5% Trimmed Mean		67.2222	
		Median		68.0000	
		Variance		5.200	
		Std. Deviation		2.28035	
		Minimum		64.00	
		Maximum		70.00	
		Range		6.00	
		Interquartile Range		4.00	
		Skewness		-.405	.913
		Kurtosis		-.178	2.000
	Female	Mean		64.4000	.74833
		95% Confidence Interval for Mean	Lower Bound	62.3223	
			Upper Bound	66.4777	
		5% Trimmed Mean		64.4444	
		Median		64.0000	
		Variance		2.800	
		Std. Deviation		1.67332	
		Minimum		62.00	
		Maximum		66.00	
		Range		4.00	
		Interquartile Range		3.00	
		Skewness		-.512	.913
		Kurtosis		-.612	2.000

Descriptives

	Gender			Statistic	Std. Error
Weight in pounds	Male	Mean		142.8000	4.55412
		95% Confidence Interval for Mean	Lower Bound	130.1557	
			Upper Bound	155.4443	
		5% Trimmed Mean		143.0556	
		Median		144.0000	
		Variance		103.700	
		Std. Deviation		10.18332	
		Minimum		128.00	
		Maximum		153.00	
		Range		25.00	
		Interquartile Range		19.00	
		Skewness		-.686	.913
		Kurtosis		-.577	2.000
	Female	Mean		123.2000	3.49857
		95% Confidence Interval for Mean	Lower Bound	113.4864	
			Upper Bound	132.9136	
		5% Trimmed Mean		122.8889	
		Median		122.0000	
		Variance		61.200	
		Std. Deviation		7.82304	
		Minimum		116.00	
		Maximum		136.00	
		Range		20.00	
		Interquartile Range		13.00	
		Skewness		1.377	.913
		Kurtosis		2.100	2.000

Percentiles

			Percentiles			
		Gender	5	10	25	50
Weighted Average (Definition 1)	Height in inches	Male	64.0000	64.0000	65.0000	68.0000
		Female	62.0000	62.0000	63.0000	64.0000
	Weight in pounds	Male	128.0000	128.0000	133.0000	144.0000
		Female	116.0000	116.0000	117.0000	122.0000
Tukey's Hinges	Height in inches	Male			66.0000	68.0000
		Female			64.0000	64.0000
	Weight in pounds	Male			138.0000	144.0000
		Female			118.0000	122.0000

Percentiles

			Percentiles		
		Gender	75	90	95
Weighted Average (Definition 1)	Height in inches	Male	69.0000	.	.
		Female	66.0000	.	.
	Weight in pounds	Male	152.0000	.	.
		Female	130.0000	.	.
Tukey's Hinges	Height in inches	Male	68.0000		
		Female	66.0000		
	Weight in pounds	Male	151.0000		
		Female	124.0000		

Extreme Values[a]

	Gender			Case Number	Value
Height in inches	Male	Highest	1	4	70.00
			2	5	68.00[b]
		Lowest	1	1	64.00
			2	9	66.00
	Female	Highest	1	2	66.00
			2	10	66.00
		Lowest	1	3	62.00
			2	7	64.00[c]
Weight in pounds	Male	Highest	1	4	153.00
			2	9	151.00
		Lowest	1	1	128.00
			2	8	138.00
	Female	Highest	1	3	136.00
			2	2	124.00
		Lowest	1	6	116.00
			2	10	118.00

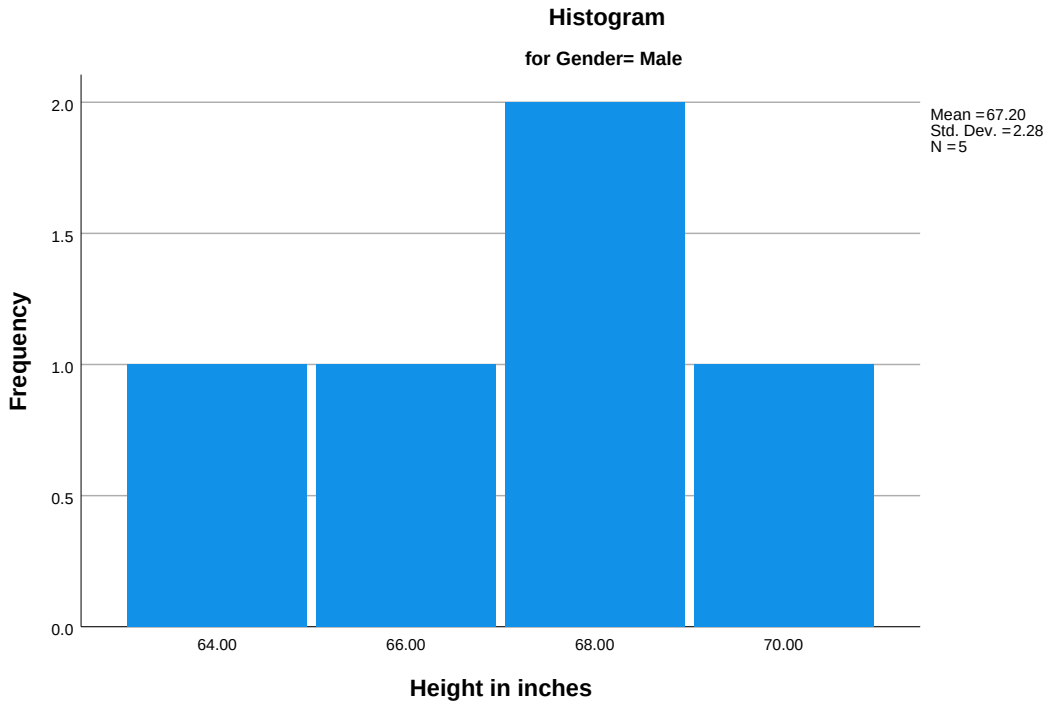
a. The requested number of extreme values exceeds the number of data points. A smaller number of extremes is displayed.

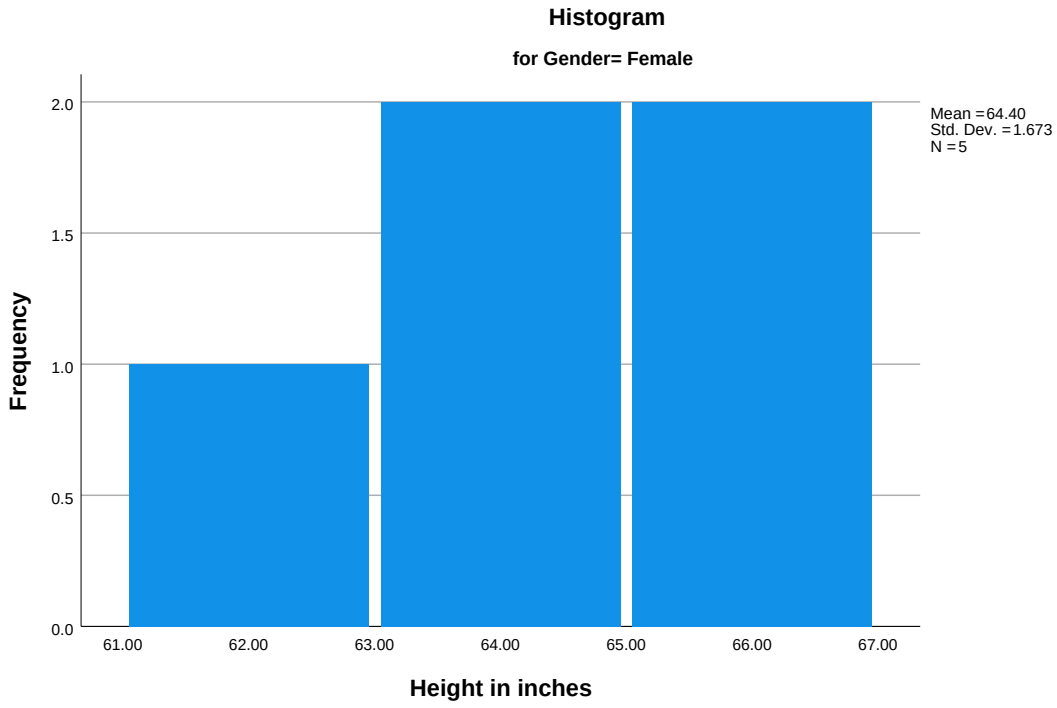
b. Only a partial list of cases with the value 68.00 are shown in the table of upper extremes.

c. Only a partial list of cases with the value 64.00 are shown in the table of lower extremes.

Height in inches

Histograms





Stem-and-Leaf Plots

```
Height in inches Stem-and-Leaf Plot for
Gender= Male
```

```
 Frequency    Stem &  Leaf

     1.00        6 .  4
     3.00        6 .  688
     1.00        7 .  0

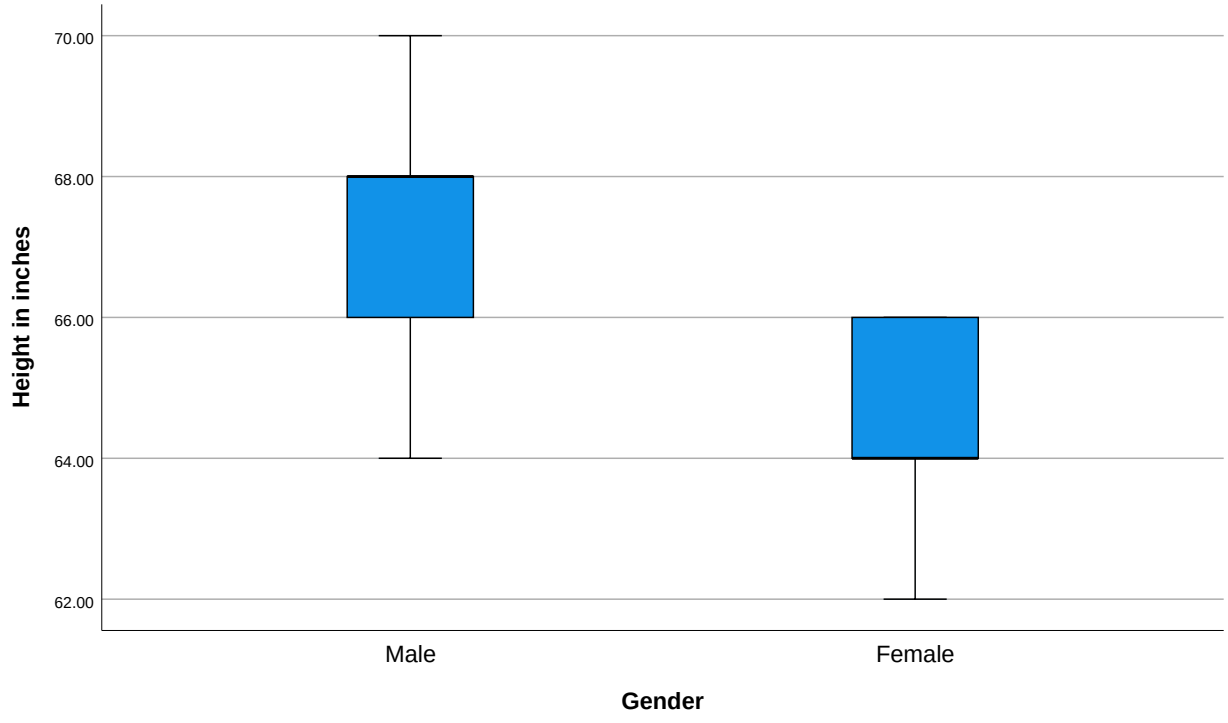
 Stem width:     10.00
 Each leaf:        1 case(s)
```

Height in inches Stem-and-Leaf Plot for
Gender= Female

```
 Frequency    Stem &  Leaf

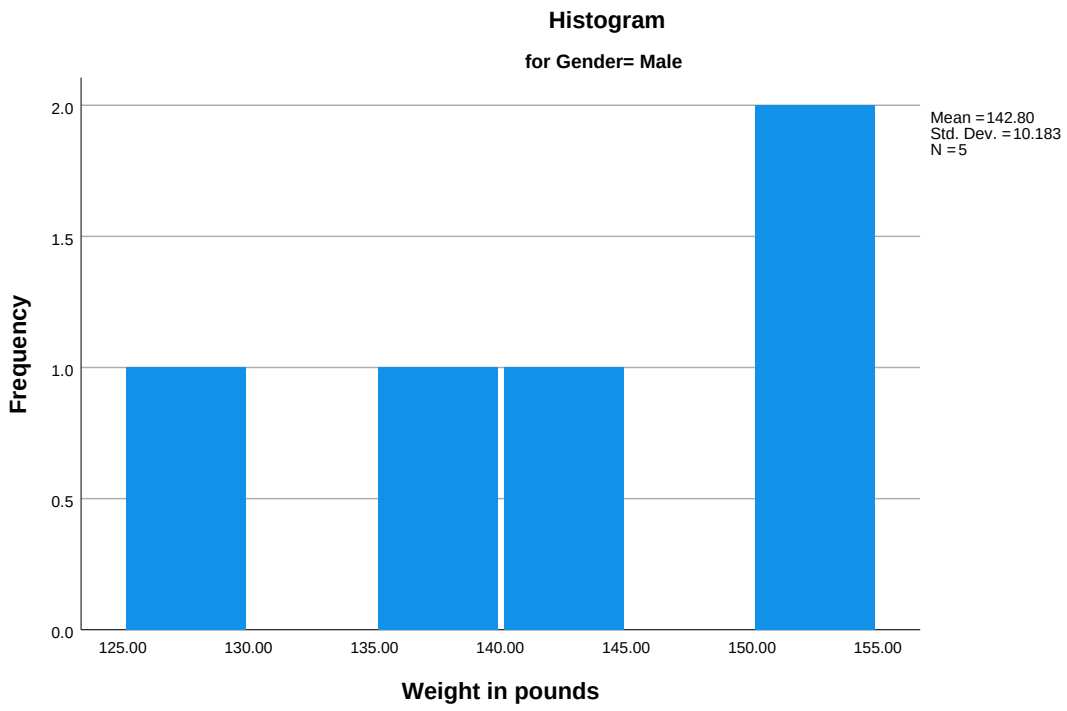
     3.00        6 .  244
     2.00        6 .  66

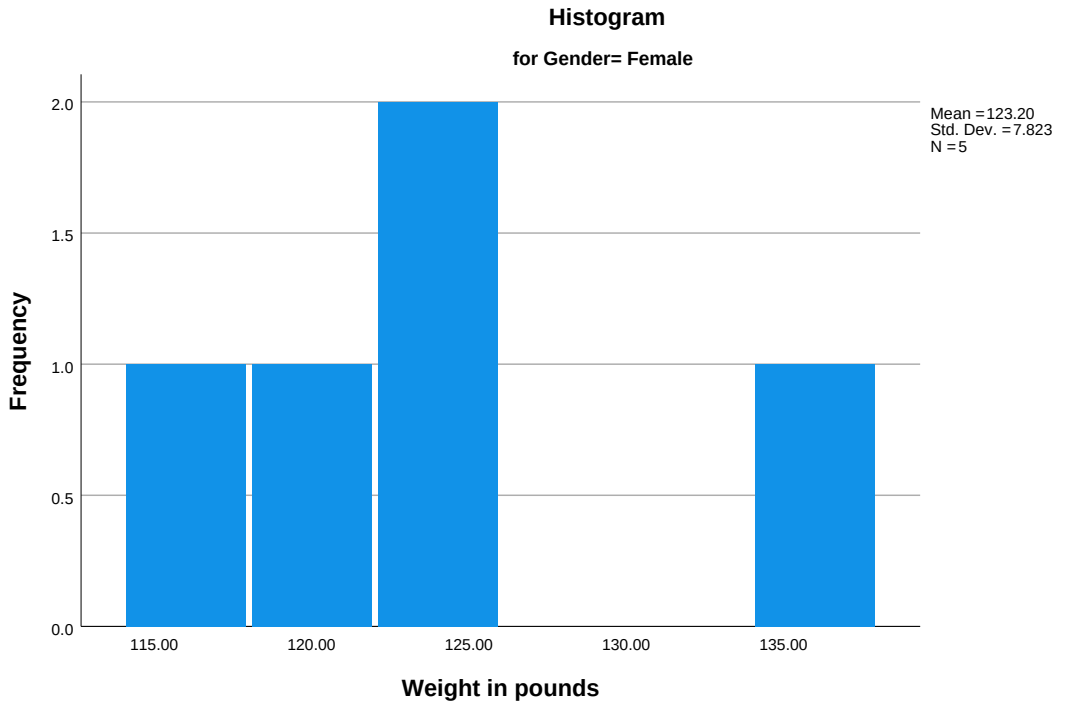
 Stem width:     10.00
 Each leaf:        1 case(s)
```



Weight in pounds

Histograms





Stem-and-Leaf Plots

```
Weight in pounds Stem-and-Leaf Plot for
Gender= Male
```

```
 Frequency    Stem &  Leaf

     1.00       12 .  8
     1.00       13 .  8
     1.00       14 .  4
     2.00       15 .  13

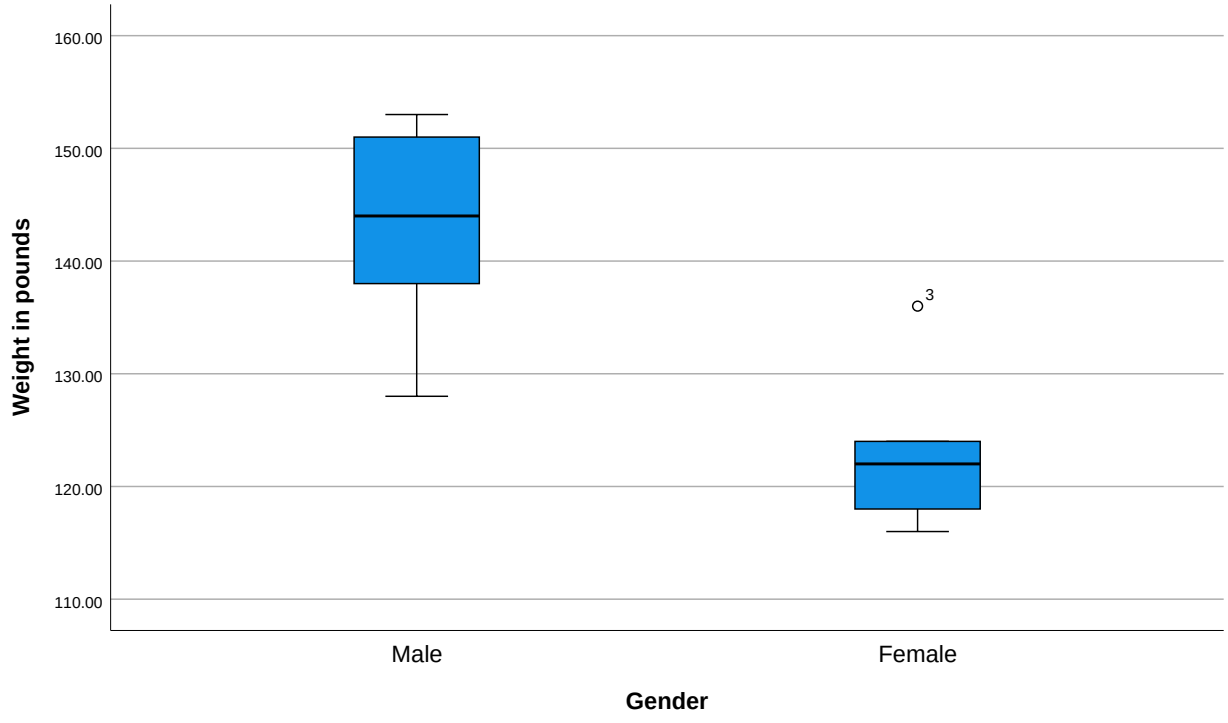
 Stem width:     10.00
 Each leaf:        1 case(s)
```

Weight in pounds Stem-and-Leaf Plot for
Gender= Female

```
 Frequency    Stem &  Leaf

     2.00       11 .  68
     2.00       12 .  24
     1.00 Extremes    (>=136)

 Stem width:     10.00
 Each leaf:        1 case(s)
```



2135 0

```
2136  0  DESCRIPTIVES VARIABLES=Height Weight
2137  0    /SAVE
2138  0    /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Notes

Output Created		03-NOV-2026 15:38:21
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Height Weight /SAVE /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.06
Variables Created or Modified	ZHeight	Zscore: Height in inches
	ZWeight	Zscore: Weight in pounds

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Height in inches	10	62.00	70.00	65.8000	2.39444
Weight in pounds	10	116.00	153.00	133.0000	13.41641
Valid N (listwise)	10				

```
2139  0
2140  0  CORRELATIONS
2141  0    /VARIABLES=Height Weight
```

```
2142  0    /PRINT=TWOTAIL NOSIG FULL
2143  0    /MISSING=PAIRWISE.
```

Correlations

Notes

Output Created		03-NOV-2026 15:38:21
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		CORRELATIONS /VARIABLES=Height Weight /PRINT=TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.09
	Elapsed Time	00:00:00.04

Correlations

		Height in inches	Weight in pounds
Height in inches	Pearson Correlation	1	.574
	Sig. (2-tailed)		.083
	N	10	10
Weight in pounds	Pearson Correlation	.574	1
	Sig. (2-tailed)	.083	
	N	10	10

```
2144  0
2145  0  CORRELATIONS
2146  0    /VARIABLES=Height Weight Gender
2147  0    /PRINT=TWOTAIL NOSIG FULL
2148  0    /MISSING=PAIRWISE.
```

Correlations

Notes

Output Created		03-NOV-2026 15:38:21
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		CORRELATIONS /VARIABLES=Height Weight Gender /PRINT=TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.09
	Elapsed Time	00:00:00.05

Correlations

		Height in inches	Weight in pounds	Gender
Height in inches	Pearson Correlation	1	.574	-.616
	Sig. (2-tailed)		.083	.058
	N	10	10	10
Weight in pounds	Pearson Correlation	.574	1	-.770**
	Sig. (2-tailed)	.083		.009
	N	10	10	10
Gender	Pearson Correlation	-.616	-.770**	1
	Sig. (2-tailed)	.058	.009	
	N	10	10	12

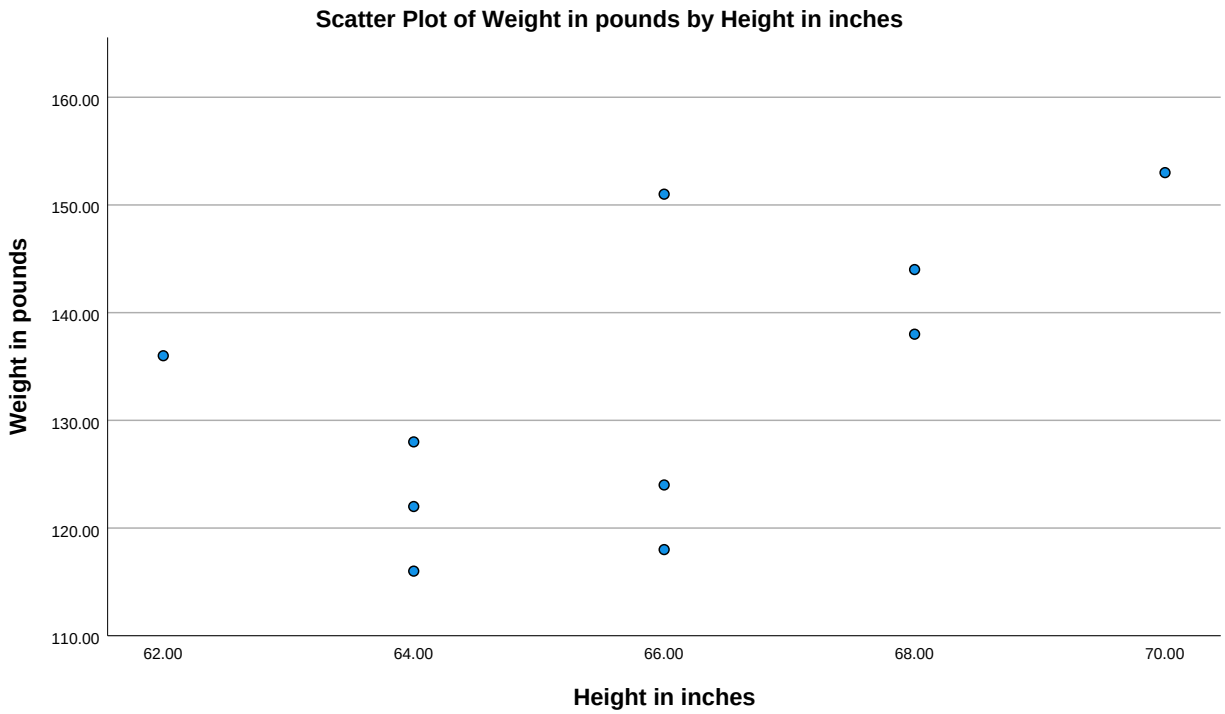
**. Correlation is significant at the 0.01 level (2-tailed).

```
2149  0
2150  0  * Chart Builder.
2151  0  GGRAPH
2152  0    /GRAPHDATASET NAME="graphdataset" VARIABLES=Height Weight MISSING=LISTWISE REP
ORTMISSING=NO
2153  0    /GRAPHSPEC SOURCE=INLINE
2154  0    /FITLINE TOTAL=NO SUBGROUP=NO.
2154  0  BEGIN GPL
2154  0    SOURCE: s=userSource(id("graphdataset"))
2154  0    DATA: Height=col(source(s), name("Height"))
2154  0    DATA: Weight=col(source(s), name("Weight"))
2154  0    GUIDE: axis(dim(1), label("Height in inches"))
2154  0    GUIDE: axis(dim(2), label("Weight in pounds"))
2154  0    GUIDE: text.title(label("Scatter Plot of Weight in pounds by Height in inches"
))
2154  0    ELEMENT: point(position(Height*Weight))
2154  0  END GPL.
```

GGraph

Notes

Output Created		03-NOV-2026 15:38:21
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Syntax		GGRAPH /GRAPHDATASET NAME="graphdataset" VARIABLES=Height Weight MISSING=LISTWISE REPORTMISSING=NO /GRAPHSPEC SOURCE=INLINE /FITLINE TOTAL=NO SUBGROUP=NO. BEGIN GPL SOURCE: s=userSource(id("graphdataset")) DATA: Height=col(source(s), name("Height")) DATA: Weight=col(source(s), name("Weight")) GUIDE: axis(dim(1), label("Height in inches")) GUIDE: axis(dim(2), label("Weight in pounds")) GUIDE: text.title(label("Scatter Plot of Weight in pounds by Height in inches")) ELEMENT: point(position(Height*Weight)) END GPL.
Resources	Processor Time	00:00:02.52
	Elapsed Time	00:00:01.08




```
2155  0
2156  0  T-TEST
2157  0    /TESTVAL=65
2158  0    /MISSING=ANALYSIS
2159  0    /VARIABLES=Height
2160  0    /ES DISPLAY(TRUE)
2161  0    /CRITERIA=CI(.95).
```

T-Test

Notes

Output Created		03-NOV-2026 15:38:22
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.
Syntax		T-TEST /TESTVAL=65 /MISSING=ANALYSIS /VARIABLES=Height /ES DISPLAY(TRUE) /CRITERIA=CI(.95).
Resources	Processor Time	00:00:00.05
	Elapsed Time	00:00:00.03

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Height in inches	10	65.8000	2.39444	.75719

One-Sample Test

	Test Value = 65					
					95% Confidence Interval of the Difference	
	t	df	Sig. (2-tailed)	Mean Difference	Lower	Upper
Height in inches	1.057	9	.318	.80000	-.9129	2.5129

One-Sample Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Height in inches	Cohen's d	2.39444	.334	-.313	.964
	Hedges' correction	2.62009	.305	-.286	.881

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation.
Hedges' correction uses the sample standard deviation, plus a correction factor.

```
2162  0
2163  0  T-TEST PAIRS=Height WITH Weight (PAIRED)
2164  0    /ES DISPLAY(TRUE) STANDARDIZER(SD)
2165  0    /CRITERIA=CI(.9500)
2166  0    /MISSING=ANALYSIS.
```

T-Test

Notes

Output Created		03-NOV-2026 15:38:22
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.
Syntax		T-TEST PAIRS=Height WITH Weight (PAIRED) /ES DISPLAY(TRUE) STANDARDIZER(SD) /CRITERIA=CI(.9500) /MISSING=ANALYSIS.

Notes

Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.02

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Height in inches	65.8000	10	2.39444	.75719
	Weight in pounds	133.0000	10	13.41641	4.24264

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	Height in inches & Weight in pounds	10	.574	.083

Paired Samples Test

		Paired Differences			
					95% Confidence ...
		Mean	Std. Deviation	Std. Error Mean	Lower
Pair 1	Height in inches - Weight in pounds	-67.20000	12.20018	3.85804	-75.92748

Paired Samples Test

		Paired ...			
		95% Confidence Interval of the ...			
		Upper	t	df	Sig. (2-tailed)
Pair 1	Height in inches - Weight in pounds	-58.47252	-17.418	9	.000

Paired Samples Effect Sizes

					95% ...
			Standardizer[a]	Point Estimate	Lower
Pair 1	Height in inches - Weight in pounds	Cohen's d	12.20018	-5.508	-8.076
		Hedges' correction	12.73976	-5.275	-7.734

Paired Samples Effect Sizes

				95% ...
				Upper
Pair 1	Height in inches - Weight in pounds	Cohen's d	-2.931	
		Hedges' correction	-2.807	

a. The denominator used in estimating the effect sizes.
Cohen's d uses the sample standard deviation of the mean difference.
Hedges' correction uses the sample standard deviation of the mean difference, plus a correction factor.

```
2167  0
2168  0  T-TEST GROUPS=Gender(1 2)
2169  0    /MISSING=ANALYSIS
2170  0    /VARIABLES=Height Weight
2171  0    /ES DISPLAY(TRUE)
2172  0    /CRITERIA=CI(.95).
```

T-Test

Notes

Output Created		03-NOV-2026 15:38:22
Comments		
Input	Data	E:\Documents\GitHub\biostatistics-portfolio\data\raw\hwg.sav
	Active Dataset	DataSet0
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	12
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on the cases with no missing or out-of-range data for any variable in the analysis.

Notes

Syntax		T-TEST GROUPS=Gender(1 2) /MISSING=ANALYSIS /VARIABLES=Height Weight /ES DISPLAY(TRUE) /CRITERIA=CI(.95).
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.02

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Height in inches	Male	5	67.2000	2.28035	1.01980
	Female	5	64.4000	1.67332	.74833
Weight in pounds	Male	5	142.8000	10.18332	4.55412
	Female	5	123.2000	7.82304	3.49857

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means	
		F	Sig.	t	df
Height in inches	Equal variances assumed	.554	.478	2.214	8
	Equal variances not assumed			2.214	7.339
Weight in pounds	Equal variances assumed	.564	.474	3.413	8
	Equal variances not assumed			3.413	7.502

Independent Samples Test

		t-test for Equality of Means		
		Sig. (2-tailed)	Mean Difference	Std. Error Difference
Height in inches	Equal variances assumed	.058	2.80000	1.26491
	Equal variances not assumed	.061	2.80000	1.26491
Weight in pounds	Equal variances assumed	.009	19.60000	5.74282
	Equal variances not assumed	.010	19.60000	5.74282

Independent Samples Test

		t-test for Equality of Means	
		95% Confidence Interval of the Difference	
		Lower	Upper
Height in inches	Equal variances assumed	-.11689	5.71689
	Equal variances not assumed	-.16322	5.76322
Weight in pounds	Equal variances assumed	6.35703	32.84297
	Equal variances not assumed	6.20234	32.99766

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval	
				Lower	Upper
Height in inches	Cohen's d	2.00000	1.400	-.044	2.778
	Hedges' correction	2.21557	1.264	-.040	2.508
	Glass's delta	1.67332	1.673	-.046	3.292
Weight in pounds	Cohen's d	9.08020	2.159	.503	3.738
	Hedges' correction	10.05890	1.949	.454	3.374
	Glass's delta	7.82304	2.505	.386	4.542

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control group.

```
2173  0
2174  0  * End of INSERT and INCLUDE nesting level 02.
2174  0
2175  0  OUTPUT EXPORT /CONTENTS EXPORT=ALL /PDF DOCUMENTFILE='E:/Documents/GitHub/biosta
tistics-portfolio/group-comparisons/spss/sp
           ss-files/hwg-ttest-correlation.pdf'.
```